

Deltares

Enabling Delta Life



EHY_tools

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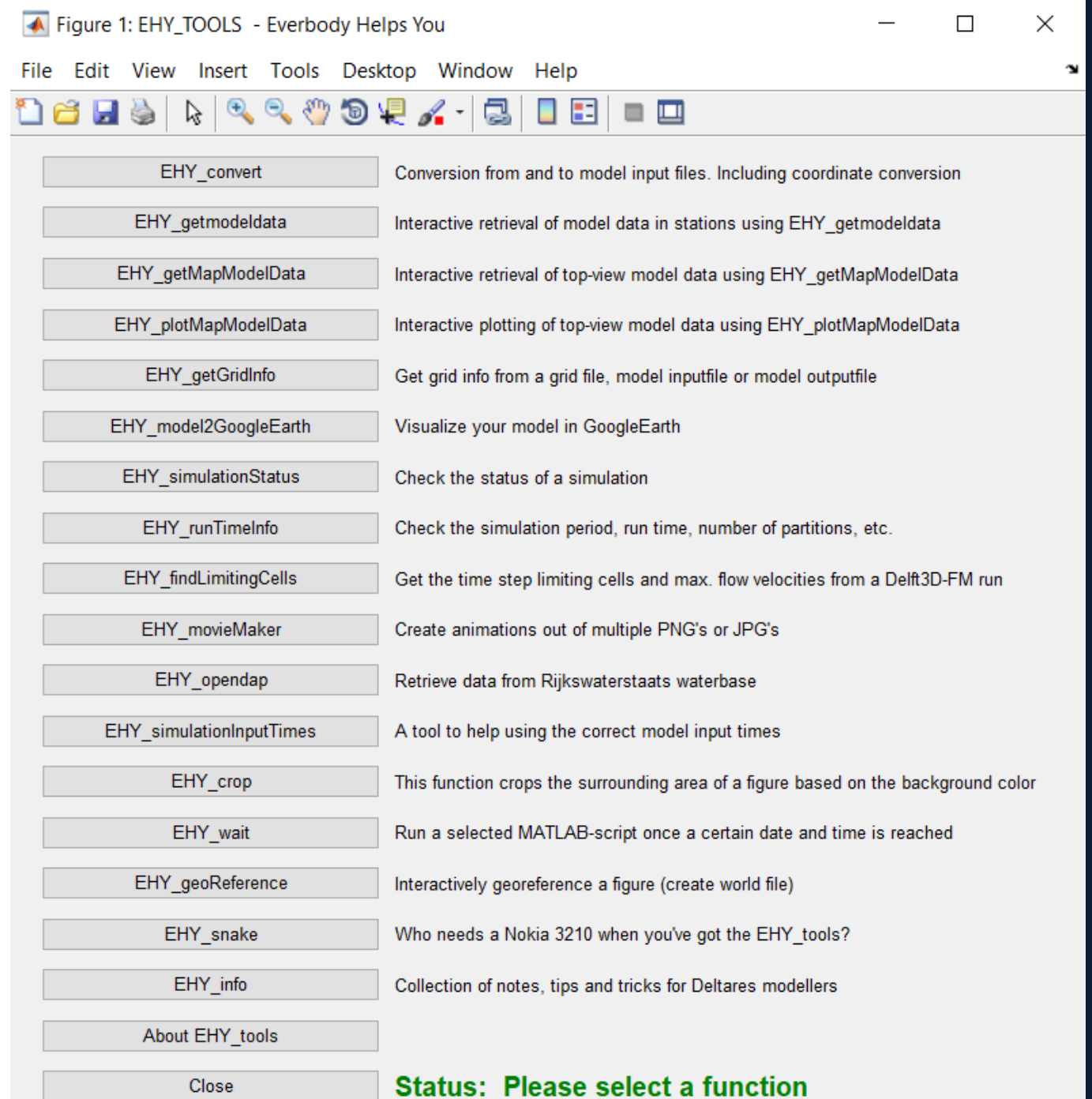
Deltares – January 2020

EHY_tools

- The **EHY_tools** is a toolbox that aims to help users of Delft3D FM, Delft3D 4, DELWAQ, SIMONA and SOBEK models in pre- and post-processing of model files using MATLAB.
- The toolbox was initially set-up and used within the group of Environmental Hydrodynamics (**EHY**) of Deltares, but is now a tool where everyone can benefit from and contribute to as well (OpenEarthTools philosophy). Therefore, the **EHY** toolbox is now short for **Everybody Helps You**.
- The aim is to have a maintained and reliable toolbox. This is pursued by checking the version-controlled scripts for erroneous modifications, using a test bench to ensure correct behaviour in future revisions and by expanding the group of users.

EHY_tools – GUI and scripts

- The scripts are created in such a way that they can be used interactively. There is a graphical user interface (GUI) that even allows non-experienced MATLAB users to make use of the available tools.
- More experienced MATLAB-users can use the functions with input and output arguments to adopt the functions in their own scripts.
- The functions contain header information with a few example lines



EHY_tools and other tools

- The EHY_tools are part of the OpenEarthTools:
`.../openearthtools/trunk/matlab/applications/EHY_tools`
- Where possible, the EHY_tools make use of existing low-level functions to avoid overlap, but it is both faster and much more flexible with respect to complete-post-processing-packages
- Compared to common post-processing approaches, the EHY_tools bridge the gap between scripting everything yourself and making plots with e.g. QuickPlot or MUPPET

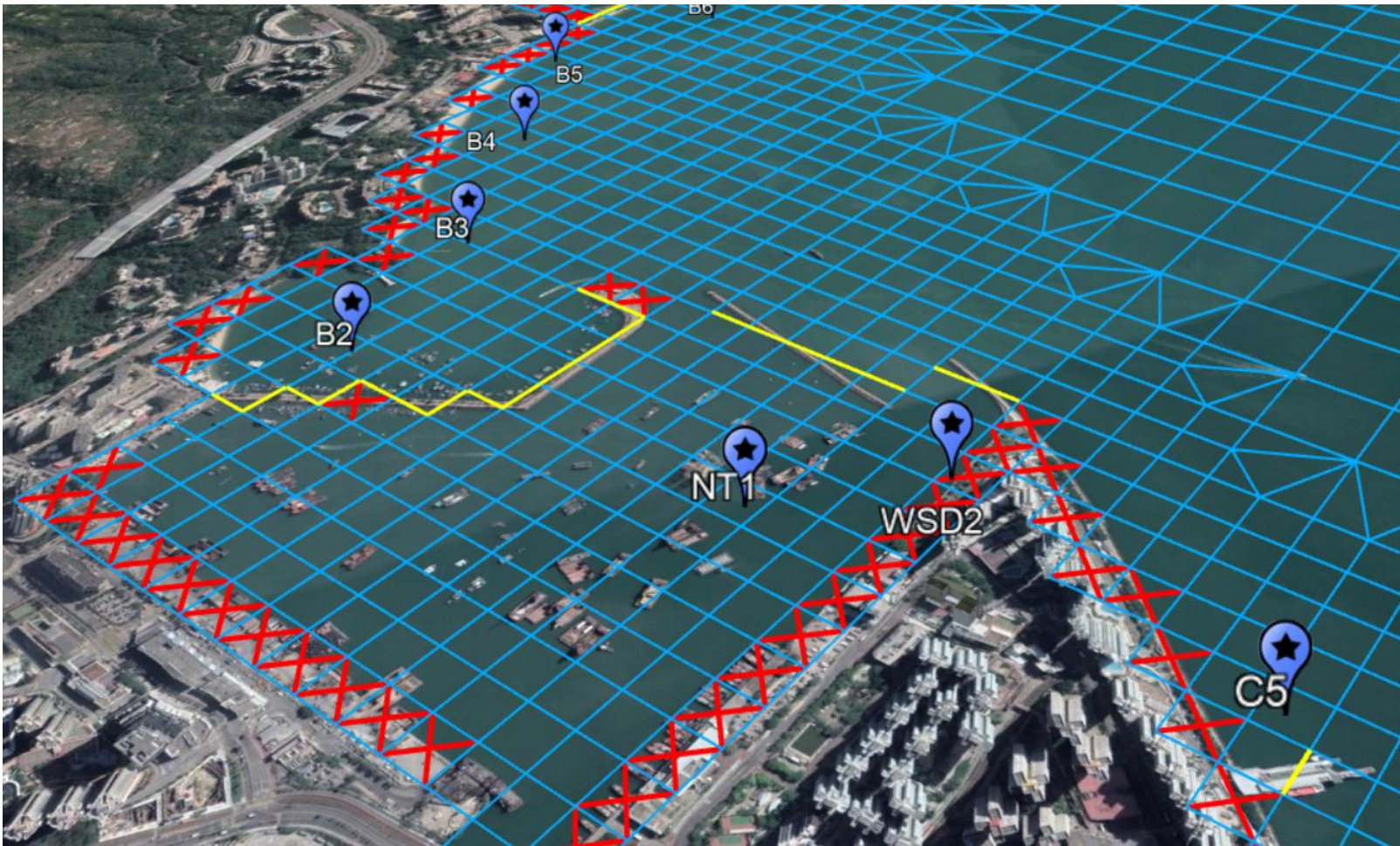


Some examples: EHY_convert

- There are many separate conversion scripts available in the OpenEarthTools. However, they all work in a slightly different way...
- EHY_convert can be seen as a wrapper that handles the correct input of these conversions (and introduces many more conversions). With just a few clicks you can convert a model file to different type of model file (e.g. obs2xyn, pol2kml and grd2shp). There are more than 50 conversions available!
- Example workflow (obs2kml):
 - EHY > EHY_convert > Open a file ... > select 'r01.obs' (Delft3D 4 observation points)
 - User is asked to convert it to: .kml, .locaties (for SIMONA) or .xyn
 - When .kml is selected, the path of the corresponding grid (.grd) is asked
 - A check of the input coordinate system is done – RDNAP and WGS'84 are automatically recognized
 - r01.obs is converted to r01.kml
 - Expert option: EHY_convert('D:\r01.obs','kml','fromEPSG',28992,'grdFile','D:\NL.grd','overwrite',1)

Some examples: EHY_model2GoogleEarth

- Combining all the conversions from EHY_convert led to EHY_model2GoogleEarth: based on a .mdu or .mdf, an entire model schematization can be converted to .kml-format with just a few clicks



Legend:
Computational grid
Thin dams
Dry points
Observation points

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Some examples: EHY_getmodeldata (1/2)

- Load history data in observation points from a Delft3D FM, Delft3D 4, SIMONA, DELWAQ or SOBEK run
- Examples:

```
% interactive mode
```

```
EHY_getmodeldata
```



Feedback after interactive retrieval of data:

Note that next time you want to get this data, you can also use:

```
Data = EHY_getmodeldata('trih-r01.dat',{'river side'},'d3d','varName','temperature');
```

```
% load water level, two stations, all times
```

```
Data = EHY_getmodeldata('trih-r01.dat',{'station1','station2'},'d3d','varName','waterlevel');
```

```
% load velocities, all stations, one year period, layer 5
```

```
Data = EHY_getmodeldata('r01_his.nc',[],'dfm','varName','uv','t0','01-Jan-2018','tend','01-Jan-2019','layer',5);
```

```
Data = EHY_getmodeldata('r01_his.nc',[],[],'varName','uv','t',1:366,'k',5);
```

Some examples: EHY_getmodeldata (2/2)

- Load history data in observation points from a Delft3D FM, Delft3D 4, SIMONA, DELWAQ or SOBEK run
- Examples:

```
% load salinity from 3D run, 1m above the bed
```

```
Data = EHY_getmodeldata('r01_his.nc','station3','dfm','varName','uv','t',1:366,'z',1,'zRef','bed');
```

```
% load salinity from 3D run, 1m below the surface
```

```
Data = EHY_getmodeldata('r01_his.nc','station3','dfm','varName','uv','t',1:366,'z',-1,'zRef','wl');
```

Example: temperature at 5m below surface



Some examples: EHY_plotMapModelData (1/3)

- Load map data from Delft3D FM, Delft3D 4 or DELWAQ run
- Interactive retrieval using EHY_plotMapModelData returns a MATLAB-line to adopt in your script
- Correct handling of deleting ghost-cells (at open boundaries and in case of multiple partitions)
- Examples:

```
% load temperature from Delft3D run, part of the grid, few time indices
```

```
m = 0; n = 10:50;
```

```
Data = EHY_getMapModelData('trim-r01.dat', 'varName', 'tem', 'm', m, 'n', n, 't', 5:10);
```

```
gridInfo = EHY_getGridInfo('trim-r01.dat', {'XYcor'}, 'm', m, 'n', n);
```

```
EHY_plotMapModelData(gridInfo, Data.val(2, :, :));
```

```
% load from partitioned Delft3D FM run
```

```
Data = EHY_getMapModelData('RunA_0002_map.nc', 'varName', 'sal', 'layer', '35'); % default: 'mergePartitions' = 1
```

```
gridInfo = EHY_getGridInfo('RunA_0002_map.nc', {'face_nodes_xy'}); % default: 'mergePartitions' = 1
```

```
EHY_plotMapModelData(gridInfo, Data.val(1, :));
```

Some examples: EHY_plotMapModelData (2/3)

- In 3D-models, Delft3D 4 and -FM (both Z- and sigma-layers), it is also possible to get the model data from:
 - A specific model layer
 - At a certain reference level (1 m below water level, 0.1 m above bed level)
 - A vertical slice (along an arbitrary cross-section) (also partitioned FM-simulations!)
- Examples:

```
% Load and plot data along cross-section
```

```
[Data,gridInfo] = EHY_getMapModelData('RunA_0002_map.nc','varName','salinity','pliFile','traject.pli');
```

```
t = 4;
```

```
EHY_plotMapModelData(gridInfo,Data.val(t,:,:),'t',t);
```

```
% Load and plot data at 1m above bed
```

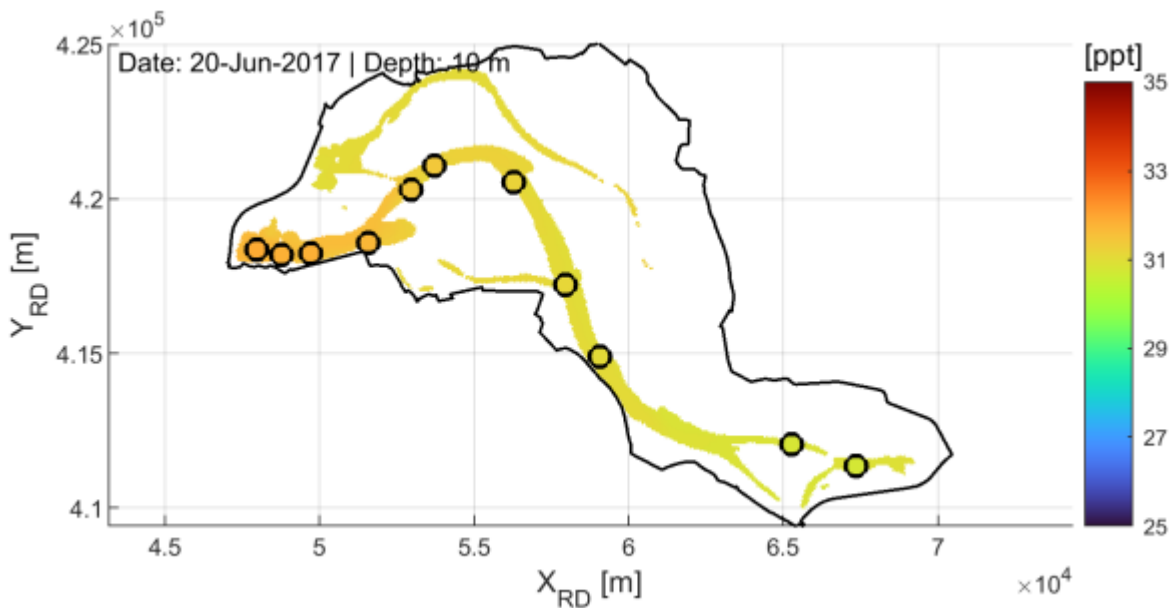
```
Data = EHY_getMapModelData('RunA_0002_map.nc','varName','temperature','zRef','bed','z','1');
```

```
gridInfo = EHY_getGridInfo('RunA_0002_map.nc',{'face_nodes_xy'});
```

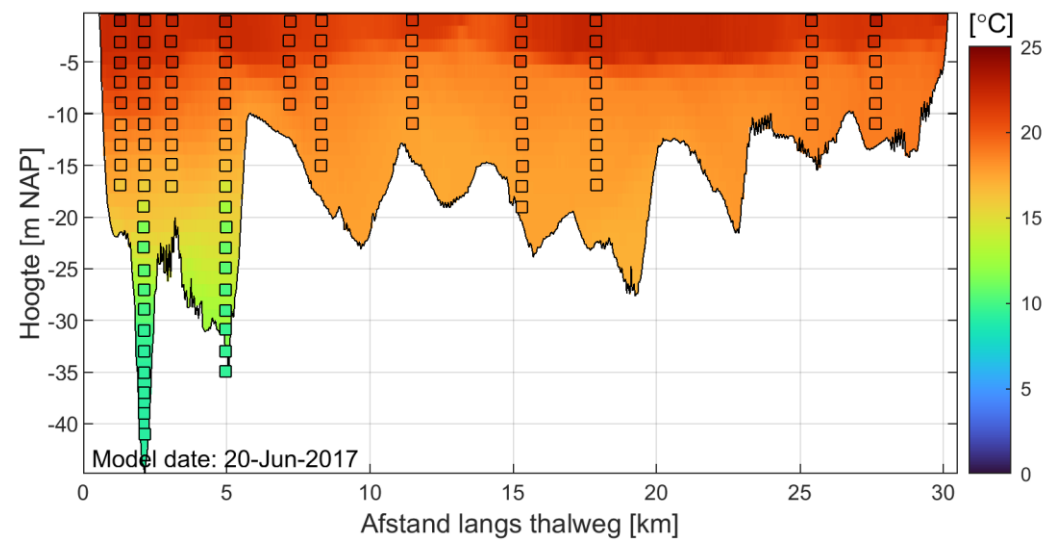
```
EHY_plotMapModelData(gridInfo,Data.val(1,:));
```

Some examples: EHY_plotMapModelData (3/3)

Example: salinity at 10m below surface



Example: temperature along arbitrary cross-section



Some examples: EHY_runTimeInfo and EHY_simulationStatus

- Example: `runTimeInfo = EHY_runTimeInfo('d:\run01\Grevelingen-FM.mdu');`

returns when finished:

struct with fields:

```
mdFile: 'd:\run01\Grevelingen-FM.mdu'
mdName: 'Grevelingen-FM.mdu'
noLayers: 36
startDate: '01-Nov-2007'
endDate: '01-Jan-2009'
simPeriod_S: 36892800
simPeriod_M: 614880
simPeriod_H: 10248
simPeriod_D: 427
simPeriod_Y: 1.1691
aveTimeStep_S: 20.6415
maxTimeStep_S: 30
realTime_S: 8.7702e+04
realTime_M: 1.4617e+03
realTime_H: 24.3618
realTime_D: 1.0151
compTime_minPerDay: 3.4232
compTime_dayPerYear: 0.8677
initTime_S: 8.8000
initTime_M: 0.1467
initextforcTime_S: 1.7280
initextforcTime_M: 0.0288
numberOfPartitions: 8
numberOfNetNodes: 23109
numberOfNetElem: 44804
```

or when still running:

struct with fields:

```
mdFile: 'd:\run02\Noordzee-FM.mdu'
mdName: 'DCSM-FM_100m.mdu'
noLayers: 1
startDate: '22-Dec-2012'
endDate: '01-Jan-2018'
simPeriod_S: 158630400
simPeriod_M: 2643840
simPeriod_H: 44064
simPeriod_D: 1836
simPeriod_Y: 5.0267
comment: 'Simulation has probably not finished yet or crashed'
status: '35.4%'
ETA: 'Simulation is expected to be finished in 1.56 days at 23-Nov-2019 21:12:45'
```

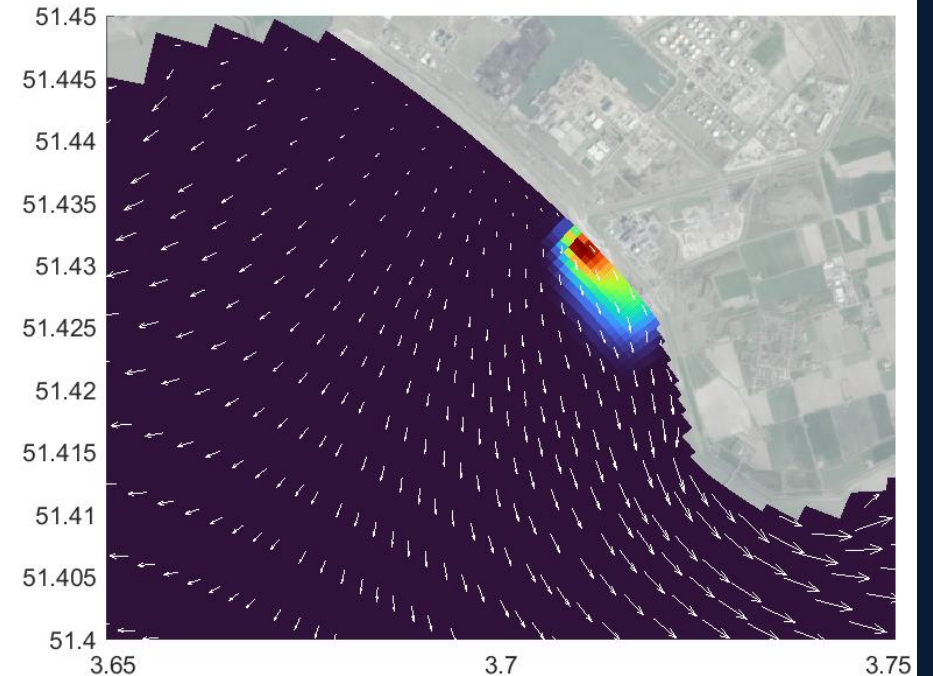
Some examples: Combination of EHY_tools-scripts

- With just a few lines, you can create nice top-view images

```
mapFile = 'r02_map.nc';

times = EHY_getmodeldata_getDatenumsFromOutputfile(mapFile);
GI = EHY_getGridInfo(mapFile,{'face_nodes_xy','XYcen','no_layers'});
topLayer = GI.no_layers;
DataTem = EHY_getMapModelData(mapFile,'varName','tem','t',5,'layer',topLayer);
DataUV = EHY_getMapModelData(mapFile,'varName','uv','t',5,'layer',topLayer);

figure
hold on
EHY_plotMapModelData(GI,DataTem.val);
EHY_quivMapModelData(GI,DataUV.vel_x,DataUV.vel_y,'thinningStyle',...
    'distance','thinning',0.002,'scaling',0.02,'color','w');
axis([3.65 3.75 51.4 51.45])
colormap(turbo)
clim([20 26])
EHY_plot_satellite_map % add satellite to background
saveas(gcf,'fig01.png')
EHY_crop('fig01.png','suffix','') % crop white border of figure
```



- Making the final adjustments (figure/axes-properties, title/labels, add other info to the plot) is up to the user.

Interested in using the EHY_tools?

- You can start using the EHY_tools by typing **EHY** in MATLAB, or have a look at the scripts in:
.../openearthtools/trunk/matlab/applications/EHY_tools (make sure to update the OET-folder on your pc)



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